

Basics of Artificial Neural Networks

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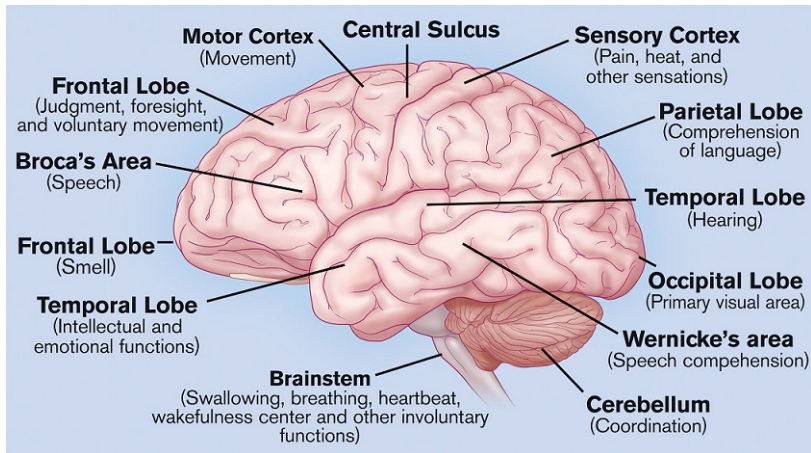


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Some Good Resources for NNDL

- Duda and Hart, "Pattern Classification" Wiley, 2001.
- C. M. Bishop, "Neural Networks for Pattern Recognition", 1995.
- B. Yegnanarayana, "Artificial Neural Networks", PHI, 1999.
- C. M. Bishop, "PRML", Springer, 2006.
- Michael Nielsen "Neural Networks and Deep Learning" Open Book.
- Ian Goodfellow, et. al. , "Deep Learning", MIT Press, 2016.
- Charu Agarwal, "NNDL", Springer 2018.
- Mithesh M. Khapra, "Deep Learning Part I and II" NPTEL Course.
- Ali Ghodsi, "Deep Learning", University of Waterloo Course.
- C. M. Bishop & H. Bishop, "Deep Learning", Springer, 2024

Human Brain



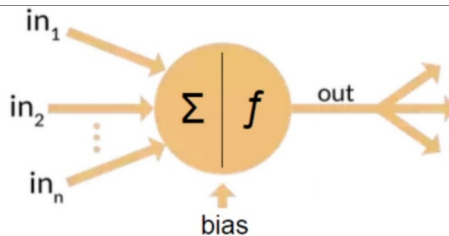
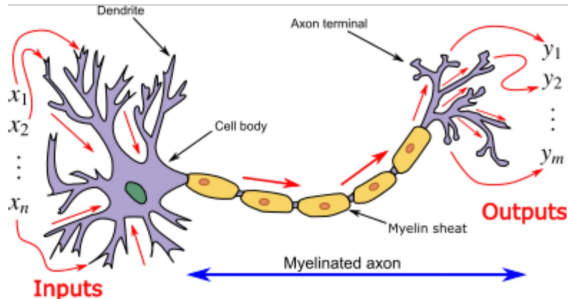
Ref.: <https://www.jworldtimes.com/old-site/magazine-archives/the-nervous-system-and-the-brain/>

Features of Biological Neural Networks

- **Robustness and fault tolerance:** Decay of nerve cells do not seem to affect performance significantly.
- **Flexibility:** Automatically adjusts to a new environment without using any preprogrammed instructions.
- **Variety of data:** Can deal with information that is fuzzy, probabilistic, noisy and inconsistent.
- **Collective computation:** Performs routinely many operations in parallel and also a given task in a distributed manner.



Biological Neuron



Biological Neuron or Nerve Cell

- Basic unit of biological NN
- **Cell body** or soma where **nucleus** located
- **Axon** long fibre extending from cell body & branches into strands connecting to many other neurons at the **synaptic junctions**.
- **Dendrites** are large no. of fibres (few thousands) receiving information from other cells
- Synaptic junctions can be on dendrites or cell body itself

Functioning of Neuron

- Transmission of a signal, a complex chemical process, specific transmitter substances are released at synaptic junctions.
- Effect is to raise or lower electrical potential inside the body of receiving cell.
- Potential reaches a threshold, an electrical activity in the form of short pulses is generated, cell is said to have fired.
- Electrical signals of fixed strength and duration sent down the axon.



Take Home Message

- Cell body acts like a **summing device**.
- Synaptic jn, which repeatedly triggers the postsynaptic neuron, **will grow in strength**, while others gradually weaken.
- **Strength** of a synaptic connection **gets modified** continuously.
- **Synaptic plasticity** in the structure of neural connectivity play a dominant role in the process of learning.
- Each neuron receiving input from several and sending out to several.
- Stacked in several layers.



Neural Networks: Terminologies

- **Processing Unit or Neuron:** Summing part + Output part.
- **Summing part** $\implies$ weighted sum with bias (activation value).
- **Output part** $\implies$ a signal from activation value using a function.
- **Layers:** Orderly arrangement of neurons according to some topology.
- **Interconnections:** Connections among different Neurons.
- **Neural Network:** Layers of interconnected neurons.



Neural Networks: Terminologies

- Activation value vs activation state vs activation function.
- Output value vs output state.
- **Activation Dynamics:** Activation state vs time.
- **Synaptic Dynamics:** Weight state (vector) vs time
- **Learning** Process of adjusting weights. Procedure to incrementally update the weights is called a **learning law or learning algorithm**.



Models of Neuron: McCulloch Pitts Model

- **Activation value** $x = \sum_{i=1}^M w_i a_i - \theta$ is a weighted summation of M inputs a_i and bias term θ .
- **Bias** (θ) allows shifting the activation function by adding a constant to the input.
- **Output Value** $s = f(x)$ is typically non-linear function (binary) of x .
- Limitation is fixed weights and no learning.
- **Capturing neuron function.**



Models of Neuron: Rosenblatt's Perceptron Model

- **Activation value** $x = \sum_{i=1}^M w_i a_i - \theta$ is a weighted summation of M inputs a_i and bias term θ .
- **Output** $s = f(x)$ is typically non-linear function (binary) of x .
- **Error** is $\delta = b - s$, where b is required / target output.
- **Weight update** is $\Delta w_i = \eta \delta a_i$, where η is learning rate parameter.
- Thus perceptron exhibits learning.
- **Breakthrough of introducing learning to a model.**
- Learning weights vs fixed weights.



Models of Neuron: Widrow's Adaline Model

- ADAPtive LINear Element (**ADALINE**) is a computing model.
- **Activation** $x = \sum_{i=1}^M w_i a_i - \theta$ is a wt. sum of M inputs a_i and θ .
- **Output** $s = f(x) = x$ is a linear function of x .
- Error is $\delta = b - s = b - x$
- Weight update is $\Delta w_i = \eta \delta a_i$.
- Weight update minimizes mean square error δ^2 .
- It is also called least mean square (LMS) error.

Topology

- Neural networks are useful only when they are organized in suitable manner (**topology**) to achieve desirable PR task.
- Units of a layer all have the **same** activation dynamics and output function.
- **Connections:** Interlayer, intralayer or both. Feedforward, feedback or both.



Learning Laws

- Activation dynamics is activation state as a function of time.
- Synaptic dynamics is weights state as a function of time.
- Learning laws are implementation of synaptic dynamics.
- $w_i(t+1) = w_i(t) + \Delta w_i(t)$, where $\Delta w_i(t)$ is change in weight at t .
- Hebb's or Hebbian law, perceptron learning law, delta learning law, Widrow and Hoff LMS learning law, correlation learning law, instar learning law, outstar learning law.



Perceptron Learning Law or Discrete perceptron law

- $\Delta \mathbf{w}_i = \eta [b_i - \text{sgn}(\mathbf{w}_i^T \mathbf{a})] \mathbf{a}$
- $\Delta w_{ij} = \eta [b_i - \text{sgn}(\mathbf{w}_i^T \mathbf{a})] a_j$, where $j = 1, 2, \dots, M$
- $\text{sgn}(x) = +1$ if $x > 0$, $= -1$ if $x < 0$ and $\text{sgn}(x) = 0$ if $x = 0$.
- Step function $f(x) = 1$, if $x > 0$ and $f(x) = 0$, if $x \leq 0$
- $\Delta w_{ij} = \eta [b_i - s_i] a_j$, where s_i is the output of i^{th} unit.
- Weights are adjusted only if the actual output s_i is incorrect.
- Weight initialization to any random values prior to learning.
- **Supervised learning** as it requires a desired output for each input.
- **Weights converge to final values by repeated use of input-output.**
- $\eta \ll 1$ due to error correction and repeated use of training patterns.

Delta Learning Law or Continuous Perceptron Law

- $\Delta \mathbf{w}_i = \eta [b_i - f(\mathbf{w}_i^T \mathbf{a})] f'(\mathbf{w}_i^T \mathbf{a}) \mathbf{a}$
- $\Delta w_{ij} = \eta [b_i - f(\mathbf{w}_i^T \mathbf{a})] f'(\mathbf{w}_i^T \mathbf{a}) a_j$, where $j = 1, 2, \dots, M$
- where $f'(x)$ is derivative wrt to x .
- $\Delta w_{ij} = \eta [b_i - s_i] f'(x) a_j$, where s_i is the output of i^{th} unit.
- **Law is valid only for differentiable output function.**
- Weight initialization to any random values prior to learning.
- **Supervised learning** as it law requires a desired output for each input.
- Weights converge to final values by repeated use of input-output.
- $\eta \ll 1$ due to error correction and repeated use of training patterns.



Widrow and Hoff LMS Learning Law

- $\Delta \mathbf{w}_i = \eta [b_i - \mathbf{w}_i^T \mathbf{a}] \mathbf{a}$
- $\Delta w_{ij} = \eta [b_i - \mathbf{w}_i^T \mathbf{a}] a_j$, where $j = 1, 2, \dots, M$
- A special case of delta learning law, where the output function is assumed linear, i.e., $f(x_i) = x_i$.
- Weight initialization to any random values prior to learning.
- **Supervised learning** as it law requires a desired output for each input.
- Weights converge to final values by repeated use of the input-output pattern pairs, in most cases. Convergence not ensured for all cases.
- $\eta \ll 1$ due to error correction and repeated use of training patterns.

